

TP1 Validator Test Cases

Team 4 CSE 360 TP1 Covers: New Username Recognizer, Password Recognizer, Email Address Validator Activity

1. Username Recognizer

State machine summary: Start → State 1 requires an ALChar (A–Z, a–z). From State 1, UNChar (A–Z, a–z, 0–9) loops back to State 1. A special character (-, _, ., &) moves State 1 → State 2, and State 2 requires a UNChar to return to State 1. A valid username must end in State 1 (i.e., never end on a special character, and never have two special characters in a row).

ID	Input	Expected Result	Rule Being Tested
TC-U01	abc	Valid	Letters only
TC-U02	abc123	Valid	Letters followed by digits
TC-U03	a1b2c3	Valid	Mixed alphanumeric
TC-U04	user-name	Valid	Single dash between two UNChars
TC-U05	user_name	Valid	Single underscore between two UNChars
TC-U06	user.name	Valid	Single period between two UNChars
TC-U07	user&name	Valid	Single ampersand between two UNChars
TC-U08	a	Valid	Minimal one-letter username

ID	Input	Expected Result	Rule Being Tested
TC-U09	1username	Invalid	Must start with ALChar, not a digit
TC-U10	-username	Invalid	Must start with ALChar, not a special character
TC-U11	user--name	Invalid	Two consecutive special characters not allowed
TC-U12	user.-name	Invalid	Two different special characters back-to-back not allowed
TC-U13	username-	Invalid	Cannot end on a special character (never returns to accepting state)
TC-U14	`` (empty string)	Invalid	No initial ALChar to consume
TC-U15	user name	Invalid	Space is not ALChar, UNChar, or an allowed special character
TC-U16	user@name	Invalid	@ is not an allowed special character for usernames

2. Password Recognizer

State machine summary: Each input character is classified as upperCase (A–Z), lowerCase (a–z), numericChar (0–9), specialChar (one of ~ ` ! @ # \$ % ^ & * () _ - + { } [] \ : , . ? /), or otherChar (anything else). charCounter increments per character. On full input consumption: longEnough = True if $8 \leq \text{charCounter} \leq 32$; tooLong = True if $\text{charCounter} > 32$.

ID	Input	Expected Result	Rule Being Tested
TC-P01	Passw0rd! (9 chars)	Valid — upperCase, lowerCase, numericChar, specialChar all True; longEnough True	All four character classes present, valid length

ID	Input	Expected Result	Rule Being Tested
TC- P02	8-character password, e.g. Aa1!Aa1!	longEnough True	Lower boundary of length range
TC- P03	7-character password, e.g. Aa1!Aa1	longEnough False	Just below minimum length
TC- P04	32-character password with mixed classes	longEnough True	Upper boundary of length range
TC- P05	33-character password with mixed classes	tooLong True	Just above maximum length
TC- P06	` ` (empty string)	charCounter = 0; all flags False; longEnough False	No input consumed
TC- P07	alllowercase	lowerCase True only (assuming length valid)	Only-lowercase input
TC- P08	ALLUPPERCASE	upperCase True only	Only-uppercase input
TC- P09	12345678	numericChar True only	Only-digit input
TC- P10	!@#\$%^&*	specialChar True only	Only allowed-special-character input
TC- P11	Password! (no digit)	numericChar False; upperCase, lowerCase, specialChar True	Missing one character class (feedback check for US-06)
TC- P12	Password containing é or a control character	otherChar True	Character outside all four defined classes
TC- P13	Password containing ; or '	otherChar True	Character not in the defined special-character list
TC- P14	Valid password with all 4 classes, length exactly 8	All relevant flags True, longEnough True, tooLong False	Combined boundary + completeness case

3. Email Address Validator

State machine summary: Local part: State 0 → State 1 on LPChar; State 1 loops on LPChar; a . moves State 1 → State 0, which requires another LPChar to return to State 1 (so a period can't lead/trail or repeat in the local part). @ moves State 1 → State 2. Domain part: State 2 → State 3 on DPChar; State 3 loops on DPChar; . moves State 3 → State 2 (needs another DPChar to return); - moves State 3 → State 4 (needs another DPChar to return to State 3). Eol (end of input) is only accepted from State 3.

ID	Input	Expected Result	Rule Being Tested
TC-E01	user@example.com	Valid	Basic well-formed address
TC-E02	first.last@example.com	Valid	Single period inside local part
TC-E03	user@my-site.com	Valid	Single dash inside domain
TC-E04	user@mail.example.co.uk	Valid	Multiple domain labels separated by periods
TC-E05	a@b.co	Valid	Minimal-length local part and domain
TC-E06	userexample.com	Invalid	Missing @
TC-E07	.user@example.com	Invalid	Local part cannot start with .
TC-E08	user..name@example.com	Invalid	Consecutive periods not allowed in local part
TC-E09	user.@example.com	Invalid	Local part cannot end on . before @

ID	Input	Expected Result	Rule Being Tested
TC-E10	@example.com	Invalid	Local part cannot be empty (no LPChar before @)
TC-E11	user@	Invalid	Domain part cannot be empty
TC-E12	user@.com	Invalid	Domain cannot start with .
TC-E13	user@example.com.	Invalid	Domain cannot end on . (EoI only valid from State 3)
TC-E14	user@example.com-	Invalid	Domain cannot end on -
TC-E15	user@exa--mple.com	Invalid	Consecutive dashes not allowed in domain
TC-E16	user@example..com	Invalid	Consecutive periods not allowed in domain
TC-E17	us er@example.com	Invalid	Space is not a valid LPChar
TC-E18	user@exam ple.com	Invalid	Space is not a valid DPChar

Prepared to support Pablo's TP1 Test Cases deliverable (target: September 15) and US-05/US-06 validation criteria per the TP1 Plan.